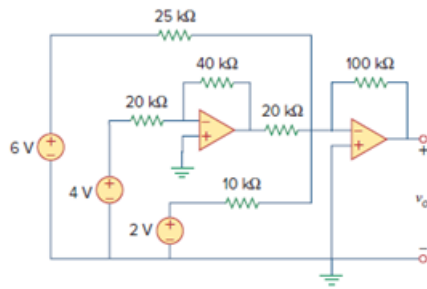


Problem

For the circuit in Fig. 5.93, find v_o .

Figure. 5.93:



Step-by-step solution

Step 1 of 3

The following is the given circuit diagram:

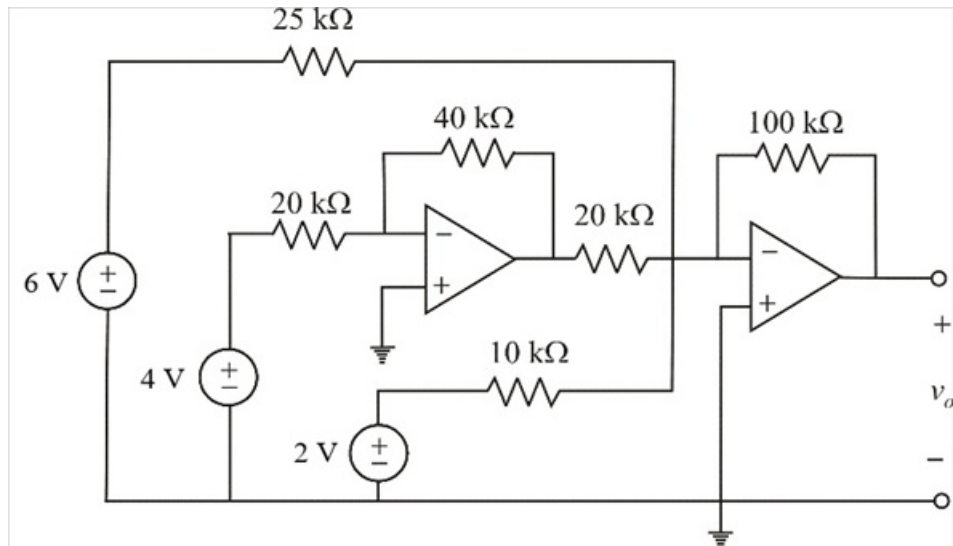


Figure 1

Step 2 of 3

This circuit consists of one inverting amplifier and one summing amplifier.

Let v_i be the output of the inverting amplifier.

Let v_o be the output of the summing amplifier.

The modified circuit with the voltages v_i is shown in Figure 2.

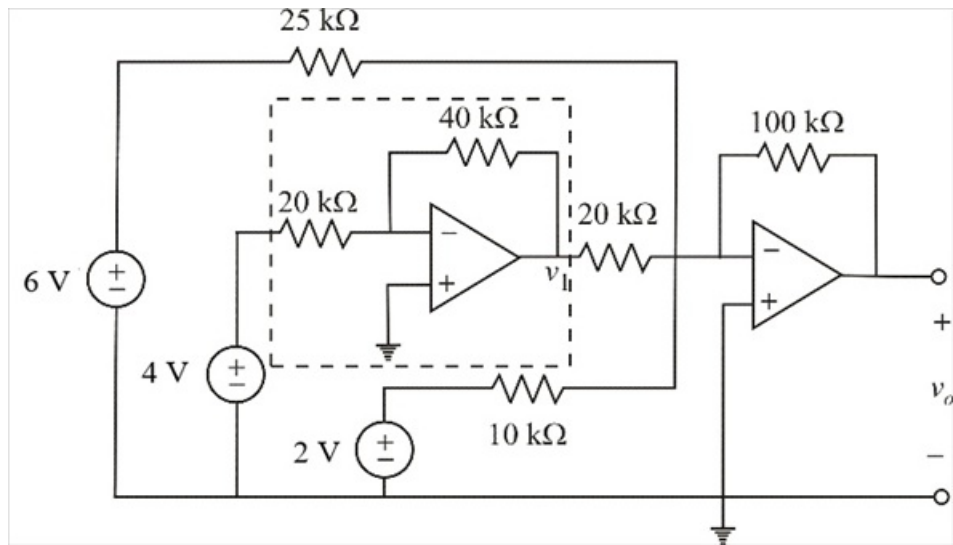


Figure 2

Step 3 of 3

Calculate the voltage v_i for the inverting amplifier output.

$$\begin{aligned} v_i &= \frac{-40\text{k}}{20\text{k}}(4) \\ &= -8 \text{ V} \end{aligned}$$

Calculate the summing amplifier output.

$$\begin{aligned} v_o &= -\left(\frac{100\text{k}}{25\text{k}}\right)(6) - \left(\frac{100\text{k}}{20\text{k}}\right)(v_i) - \left(\frac{100\text{k}}{10\text{k}}\right)(2) \\ &= -(4)(6) - (5)(-8) - (10)(2) \\ &= -24 + 40 - 20 \\ &= -4 \text{ V} \end{aligned}$$

Therefore, the output of the amplifier circuit is $\boxed{-4 \text{ V}}$.